

JEE MAIN CRASH COURSE

IT'S TIME TO LEARN APPLICATION

- ✓ LATEST NTA PATTERN
- ✓ CONCEPT VIDEOS
- ✓ LIVE DOUBT SESSIONS
- ✓ DPP, REVISION SHEETS
- ✓ 100% MONEYBACK TO MERITORIOUS STUDENTS

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Designed by
Math Maestro

Anup Sir



JEE ADV. CRASH COURSE

TARGETING JEE ADVANCED 2020

- ✓ CONCEPT BUILDING VIDEOS
- ✓ ADVANCED LEVEL VIDEOS
- ✓ 2 LEVELS OF SHEETS,
1 WITH ALL TYPE OF QUES.
- ✓ LIVE DISCUSSION OF
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JEE Mains 2020 Jan Chapter wise Question Bank

Indefinite Integration

01

Let $\int \frac{\cos x \, dx}{\sin^3 x (1 + \sin^6 x)^{\frac{2}{3}}} = f(x) (1 + \sin^6 x)^{\frac{1}{\lambda}} + c$ then find value of $\lambda f\left(\frac{\pi}{3}\right)$

माना $\int \frac{\cos x \, dx}{\sin^3 x (1 + \sin^6 x)^{\frac{2}{3}}} = f(x) (1 + \sin^6 x)^{\frac{1}{\lambda}} + c$, तो $\lambda f\left(\frac{\pi}{3}\right)$ का मान है—

(1) 4

(2) -2

(3) 8

(4) -4

8th Jan Morning

Sol

(2)

$$\sin x = t$$

$$\cos x \, dx = dt$$

$$I = \int \frac{dt}{t^3 (1 + t^6)^{\frac{2}{3}}} = \int \frac{dt}{t^7 \left(1 + \frac{1}{t^6}\right)^{\frac{2}{3}}}$$

$$\text{Put } 1 + \frac{1}{t^6} = r^3 \text{ रखने पर } \Rightarrow \frac{dt}{t^7} = \frac{-1}{2} r^2 dr$$

$$-\frac{1}{2} \int \frac{r^2 dr}{r^2} = -\frac{1}{2} r + c = -\frac{1}{2} \left(\frac{\sin^6 x + 1}{\sin^6 x} \right)^{\frac{1}{3}} + c = -\frac{1}{2 \sin^2 x} (1 + \sin^6 x)^{\frac{1}{3}} + c$$

$$f(x) = -\frac{1}{2} \operatorname{cosec}^2 x \text{ and और } \lambda = 3$$

$$\lambda f\left(\frac{\pi}{3}\right) = -2$$

02

Find integration $\int \frac{dx}{(x-3)^{6/7} \cdot (x+4)^{8/7}}$

समाकलन $\int \frac{dx}{(x-3)^{6/7} \cdot (x+4)^{8/7}} =$

- (1) $\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + c$ (2) $7\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + c$ (3) $7\left(\frac{x-3}{x+4}\right)^{\frac{6}{7}} + c$ (4) $7\left(\frac{x+4}{x-3}\right)^{\frac{6}{7}} + c$

9th Jan Morning

Sol

(1)

$$\int \left(\frac{x-3}{x+4}\right)^{\frac{-6}{7}} \frac{1}{(x+4)^2} dx$$

Let $\frac{x-3}{x+4} = t^7,$

$$\frac{7}{(x+4)^2} dx = 7t^6 dt$$

$$\int t^{-6} t^6 dt = t + c$$

03

$\int \frac{d\theta}{\cos^2 \theta (\sec 2\theta + \tan 2\theta)} = \lambda \tan \theta + 2 \log f(x) + c$ then ordered pair $(\lambda, f(x))$ is

$\int \frac{d\theta}{\cos^2 \theta (\sec 2\theta + \tan 2\theta)} = \lambda \tan \theta + 2 \log f(x) + c$ तब क्रमित युग्म $(\lambda, f(x))$ है—

- (1) $(1, 1 + \tan \theta)$ (2) $(1, 1 - \tan \theta)$ (3) $(-1, 1 + \tan \theta)$ (4) $(-1, 1 - \tan \theta)$

9th Jan Evening

Sol

(3)

$$\int \frac{\sec^2 \theta}{\frac{1+\tan^2 \theta}{1-\tan^2 \theta} + \frac{2\tan \theta}{1-\tan^2 \theta}} d\theta$$

$$= \int \frac{\sec^2 \theta (1-\tan^2 \theta)}{(1+\tan \theta)^2} d\theta$$

$$= \int \frac{\sec^2 \theta (1-\tan \theta)}{1+\tan \theta} d\theta$$

$$\tan \theta = t \Rightarrow \sec^2 \theta d\theta = dt$$

$$= \int \left(\frac{1-t}{1+t} \right) dt = \int \left(-1 + \frac{2}{1+t} \right) dt$$

$$= -t + 2 \log (1+t) + C$$

$$= -\tan \theta + 2 \log (1 + \tan \theta) + C$$

$$\Rightarrow \lambda = -1 \text{ and } f(x) = 1 + \tan \theta$$



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